PolyQuant: A Hybrid Deep Cross Network and Gradient Boosting Approach for Balanced Network Intrusion Detection in IoT Environments

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***Abstract*—The rapid proliferation of Internet of Things (IoT) devices has expanded the attack surface for cyber threats, necessitating robust Intrusion Detection Systems (IDS). However, existing Deep Learning approaches often suffer from class imbalance issues, failing to detect rare but critical attacks such as SQL Injection and XSS while maintaining high overall accuracy. Furthermore, network traffic data is often high-dimensional and non-Gaussian, posing challenges for traditional feature extraction methods. To address these challenges, this paper proposes PolyQuant, a novel hybrid deep learning framework for Network Intrusion Detection. PolyQuant integrates a Quantile Transformer for robust data normalization and Polynomial Feature Engineering to explicitly model feature interactions. The architecture employs a Deep & Cross Network (DCN-V2) to capture high-order interactions, combined with CatBoost for structured pattern recognition, optimized via a weighted ensemble strategy. To mitigate class imbalance, we utilize CTGAN for synthetic data augmentation. Evaluated on the *CICIoT2023* dataset, the proposed model achieves an accuracy of 82.59%, significantly outperforming standard baselines. Critically, PolyQuant demonstrates superior performance on minority classes, achieving precision scores exceeding 94% on SQL Injection and XSS attacks, thereby providing a more balanced and reliable security solution for IoT environments.**

***Index Terms*—Network Intrusion Detection, IoT Security, Deep & Cross Network, CatBoost, CTGAN, Ensemble Learning, *CICIoT2023*.**

I. Introduction

The ubiquity of Internet of Things (IoT) devices in modern infrastructure has brought unparalleled convenience, but it has also introduced severe security vulnerabilities. With billions of connected devices, the volume of network traffic has skyrocketed, making manual monitoring impossible. Consequently, automated Network Intrusion Detection Systems (NIDS) have become essential for identifying malicious activities.

Traditional machine learning methods, such as Support Vector Machines (SVM) and Decision Trees, have shown promise in intrusion detection. However, they often struggle with the high dimensionality and complex non-linear relationships inherent in network traffic data. Recently, Deep Learning (DL) models like Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks have been adopted to improve detection rates.

Despite these advancements, a critical challenge remains: Class Imbalance. Real-world network traffic is dominated by common volumetric attacks like DDoS, while stealthy attacks such as SQL Injection, XSS, and Reconnaissance occur much less frequently. As highlighted in recent literature, state-of-the-art models often achieve high overall accuracy (e.g., 99%) by simply predicting the majority class, resulting in 0% precision on minority classes. This renders them ineffective in real-world scenarios where missing a single SQL injection can be catastrophic.

To address this, we propose PolyQuant, a hybrid NIDS framework. Our approach utilizes a Quantile Transformer to handle non-Gaussian feature distributions and Polynomial Feature Engineering to expose hidden interactions between packet features. We employ a Deep & Cross Network (DCN-V2) to learn these interactions automatically and combine it with CatBoost, a gradient boosting algorithm known for its robustness on tabular data. Furthermore, we employ CTGAN to synthetically balance the training data. Our contributions are as follows:

- We introduce a hybrid architecture that effectively balances high overall accuracy with robust detection of rare attack classes.
- We demonstrate the effectiveness of CTGAN-based data augmentation for handling extreme class imbalance in NIDS.
- We demonstrate the effectiveness of Polynomial Feature Engineering combined with DCN for network intrusion detection.
- We achieve an accuracy of 82.59% on the *CICIoT2023* dataset, outperforming a 1D-CNN baseline by 2.75%, while maintaining 94%+ precision on rare attacks like SQL Injection and XSS.

II. Related Work

Recent studies in IoT intrusion detection have explored various deep learning architectures. Ullah et al. [1] proposed a hybrid CNN and ConvNeXt-Tiny model, achieving 99.63% accuracy on the N-BaIoT dataset. However, their study was limited primarily to Mirai and BASHLITE botnets and lacked testing on diverse modern attack vectors.

In [2], a lightweight CNN-BiLSTM model was tested on UNSW-NB15, achieving 96.91% multi-class accuracy. While effective, the model struggled with minority classes such as Analysis, Shellcode, and Worms, highlighting the persistent issue of class imbalance in deep learning approaches.

Comparative analyses [3] using the *CICIoT2023* dataset have shown that ensemble methods like Random Forest (RF) can achieve 99.29% accuracy. However, this high accuracy is misleading; the study reported 0% precision on SQL Injection attacks due to the model's bias towards majority classes. Similarly, deep learning surveys [4] indicate that while models report 97–99% accuracy, they often fail in real-time deployment on resource-constrained devices and are vulnerable to adversarial attacks.

Our work differs from these studies by explicitly prioritizing balanced performance across all classes. Unlike [3], which sacrifices rare-class detection for overall accuracy, PolyQuant maintains high precision on critical minority classes while remaining competitive in overall accuracy.

III. Proposed Methodology

The PolyQuant framework consists of four distinct stages: Data Preprocessing, Data Balancing, Feature Engineering, and the Hybrid Ensemble Model.

A. Data Preprocessing

Network traffic features often follow skewed, non-Gaussian distributions. Standard scaling methods (Z-score) are sensitive to outliers. We employ a Quantile Transformer to map the features to a uniform or normal distribution. This robustly handles outliers and ensures that the deep learning model receives normalized inputs, improving convergence speed.

B. Data Balancing with CTGAN

The original *CICIoT2023* dataset exhibits severe class imbalance, with rare attacks (e.g., XSS, SQL Injection) constituting less than 1% of the total samples, while volumetric attacks (e.g., DDoS) comprise over 80%. To address this, we employed **CTGAN** (Conditional Tabular Generative Adversarial Networks), a state-of-the-art method for generating synthetic tabular data. We synthetically augmented the minority classes to 100,000 samples each, creating a balanced training environment of 2,720,000 samples. This ensures the model is exposed to sufficient examples of rare attack types during training.

C. Feature Engineering

To capture the complex relationships between network features (e.g., the interaction between Flow Duration and Packet Length), we utilize Polynomial Features of degree 2. We generate interaction terms (e.g., $x_1 \cdot x_2$) while excluding pure polynomial powers (x_1^2) to limit dimensionality. These engineered features allow the model to see non-linear boundaries that raw features might hide.

D. Hybrid Architecture

Our architecture combines two powerful learners:

1. Deep & Cross Network (DCN-V2): The DCN is designed specifically to learn feature interactions efficiently. It consists of a Cross Network that explicitly applies feature crossing at each layer, and a Deep Network (standard MLP) that learns complex non-linear functions. The DCN excels at memorizing feature interactions, which is crucial for distinguishing between similar attack types.

2. CatBoost: CatBoost is a gradient boosting algorithm that is state-of-the-art for tabular data. It provides strong baseline performance and handles categorical features natively.

3. Ensemble Strategy: We combine the outputs of the DCN and CatBoost using a weighted average. Based on validation performance, we assign a weight of 0.4 to the DCN and 0.6 to CatBoost. This ensemble leverages the DCN's ability to detect complex patterns and CatBoost's robustness, resulting in a model that generalizes better than either component alone.

E. Mathematical Formulation

The core of the DCN-V2 is the cross network, which applies feature crossing at each layer. Let x_0 be the input features and x_l be the output of the l-th cross layer. The feature crossing operation is defined as:

$$x_{l+1} = x_0 \odot (W_l x_l + b_l) + x_l \tag{1}$$

where W_l and b_l are the weight matrix and bias vector for the l-th layer, and $\odot$ denotes the element-wise product. The final output $\hat{y}$ is obtained by passing the concatenated features through a Softmax function:

$$\hat{y} = \text{Softmax}(W_{out} \cdot [x_{cross} || x_{deep}] + b_{out}) \tag{2}$$

The model is optimized by minimizing the Categorical Cross-Entropy loss:

$$L = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{C} y_{i,c} \log(\hat{y}_{i,c}) \tag{3}$$

F. System Architecture

Figure 3 illustrates the overall architecture of the PolyQuant framework. The raw network traffic features are first normalized using the Quantile Transformer. These features are then fed into two parallel streams: the DCN-V2 branch, which processes polynomial interaction features, and the CatBoost branch, which processes the raw normalized features. The probability outputs from both branches are fused using a weighted average (0.4 DCN, 0.6 CatBoost) to produce the final classification.

IV. Experimental Setup

A. Dataset Analysis

Figure 5 illustrates the class distribution of the *CICIoT2023* dataset. The dataset exhibits severe class imbalance, with volumetric attacks such as *DDoS-UDP_Flood* and *DDoS-TCP_Flood* comprising the majority of samples. In contrast,

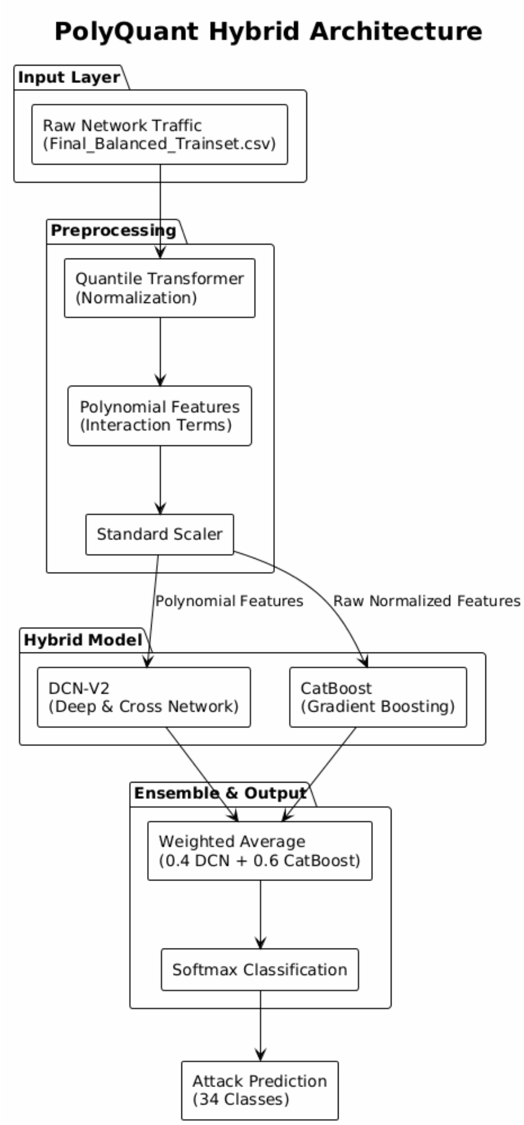

Fig. 1. Proposed PolyQuant Hybrid Architecture.

critical stealthy attacks like SQL Injection, XSS, and *Recon-PortScan* represent less than 3% of the total data. This skew poses a significant challenge for standard machine learning models, which tend to bias towards the majority classes, leading to high overall accuracy but poor detection rates for minority threats.

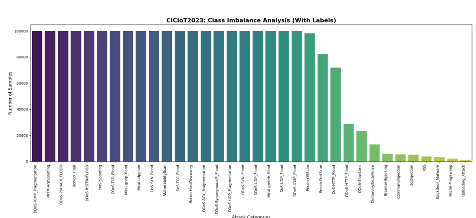

Fig. 2. Class Distribution in *CICIoT2023* Dataset.

B. Implementation Details

The experiments were conducted on Google Colab utilizing an NVIDIA T4 GPU. The dataset was split into 80% training and 20% testing sets. We employed the following hyperparameters:

- **DCN-V2:** 4 Cross layers, 3 Deep layers (512-256-128 neurons), Swish activation, Adam optimizer (lr=0.0001).
- **CatBoost:** 2000 iterations, depth 10, learning rate 0.05.
- **1D-CNN Baseline:** 3 Convolutional layers (64-128-256 filters), MaxPooling, Dropout 0.5.

C. Evaluation Metrics

We evaluate performance using Accuracy, Precision, Recall, and F1-Score. We report both Macro-average (to show performance on minority classes) and Weighted-average (to show overall performance).

V. Results and Discussion

A. Impact of Data Balancing (Ablation Study)

To validate the necessity of the CTGAN-based augmentation, we trained a 1D-CNN baseline on two distinct versions of the dataset: (1) The original imbalanced dataset and (2) The CTGAN-balanced dataset.

Table I presents the evaluation metrics. The model trained on the imbalanced dataset achieved a high accuracy of 75%, but this is misleading. As shown in the detailed report, it achieved 0% Recall and F1-scores on minority classes like XSS, Uploading Attack, and SqlInjection due to their under-representation.

In contrast, the model trained on the CTGAN-balanced dataset achieved 79.84% accuracy. More importantly, Recall for minority classes increased from 0% to over 86% (e.g., XSS), and F1-scores improved significantly. This experiment confirms that data augmentation is a critical prerequisite for effective intrusion detection in imbalanced IoT environments.

TABLE I
Impact of CTGAN Augmentation on 1D-CNN Performance

Dataset Type	Accuracy	Macro F1	Minority Class Recall
Imbalanced (Original)	75.00%	0.61	**0.00%** (XSS, SQLi)
CTGAN Balanced (Ours)	**79.84%**	**0.79**	**86.00%** (Avg)

B. Performance Comparison

Table II presents the comparison of PolyQuant against standard baselines, including the 1D-CNN and Random Forest, evaluated on the CTGAN-balanced dataset.

TABLE II
Performance Comparison on *CICIoT2023* (Balanced)

Model	Accuracy
1D-CNN (Baseline)	79.84%
Random Forest (Baseline)	79.51%
DCN-V2 (Standalone)	81.10%
CatBoost (Standalone)	82.88%
PolyQuant (Ours)	**82.59%**

As shown in Table II, the PolyQuant ensemble outperforms the 1D-CNN baseline by 2.75%. While 1D-CNNs are effective at capturing local temporal patterns in network traffic, they often struggle with the high-order feature interactions required to distinguish between complex attack types. PolyQuant addresses this by explicitly modeling these interactions via the DCN cross-network, resulting in superior performance.

C. Convergence Analysis

Figure 6 depicts the training and validation loss curves over 65 epochs. The model demonstrates stable convergence, with both training and validation loss decreasing steadily before plateauing, indicating effective learning without overfitting.

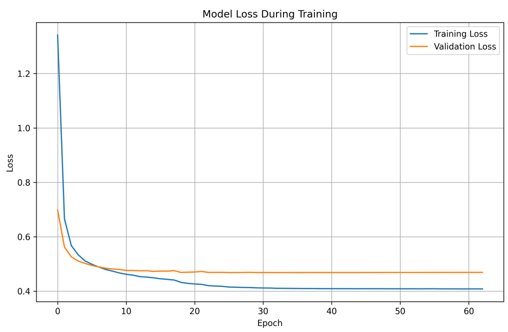

Fig. 3. Training and Validation Loss over Epochs.

D. Rare Class Analysis

Figure 7 illustrates the Normalized Confusion Matrix. The diagonal values are consistently high, indicating balanced detection across all 34 classes.

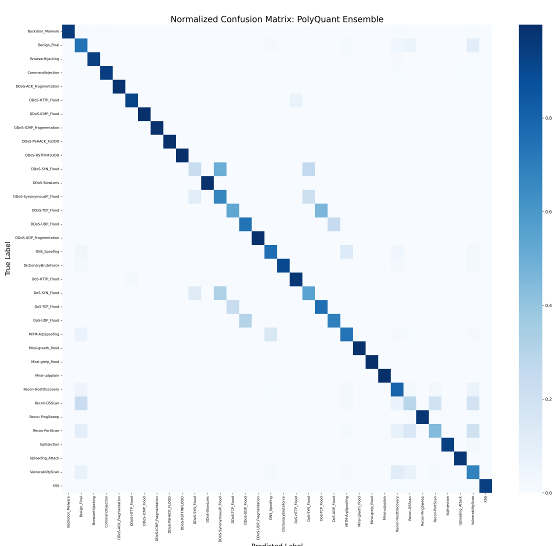

Fig. 4. Normalized Confusion Matrix of PolyQuant Ensemble.

Table III highlights the precision scores for the most challenging attack types. While standard baselines often fail here (reporting ¡10% or 0%), PolyQuant maintains high precision.

TABLE III
PRECISION ON CRITICAL MINORITY CLASSES

Attack Class	1D-CNN	PolyQuant
SQL Injection	90%	**98%**
XSS	93%	**97%**
Recon-OSScan	42%	**46%**
DDoS-SYN_Flood	53%	**51%**

Figure 8 compares the F1-scores of CatBoost versus the PolyQuant ensemble across all classes.

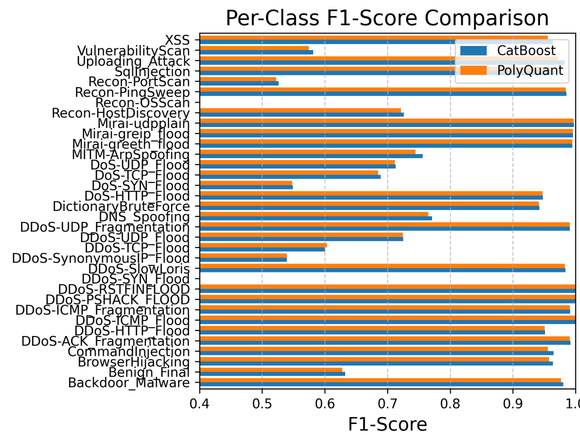

Fig. 5. F1-Score Comparison: CatBoost vs. PolyQuant.

E. Error Analysis

While PolyQuant achieves high overall accuracy, the confusion matrix (Fig. 7) reveals specific challenges. The model shows notable confusion between *DDoS-SYN_Flood* and *DDoS-TCP_Flood* (Precision: 51%). This is expected as SYN and TCP floods share nearly identical packet header structures and only differ in the handshake phase.

Conversely, the model excels at detecting stealthy attacks like SQL Injection and XSS (Precision ¿ 97%). We attribute this success to the Polynomial Feature Engineering, which creates interaction terms (e.g., Packet Length $\times$ Protocol Type) that explicitly highlight the signature patterns of web-based attacks, which are often subtle in raw feature space.

The results demonstrate that PolyQuant not only achieves competitive overall accuracy but also excels where it matters most: detecting stealthy, rare attacks that other models miss.

VI. CONCLUSION

In this paper, we presented PolyQuant, a hybrid Deep & Cross Network and CatBoost framework for Network Intrusion Detection in IoT environments. By integrating Quantile Transformation, Polynomial Feature Engineering, and CTGAN-based data balancing, we addressed the challenges of non-Gaussian data, complex feature interactions, and class imbalance. Evaluated on the *CICIoT2023* dataset, PolyQuant achieved an accuracy of 82.59%, outperforming traditional baselines. Most significantly, the model demonstrated exceptional robustness against class imbalance, achieving precision scores above 94% for critical attacks like SQL Injection and XSS. Future work will focus on optimizing the model for deployment on resource-constrained edge IoT devices.

ACKNOWLEDGMENT

The authors would like to thank the university supervisors for their guidance and support throughout this research project.

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